Doddavarapu Pavan Venkat Kumar

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Technical Skills

Languages: Java, Python, C++, C, SQL

Web & Backend: HTML, CSS, FAST APIs, MySQL

Technologies/Frameworks: LangChain, Scikit-learn, Hugging Face, Streamlit, n8n,

Tools & Fundamentals: Git, GitHub, Docker (Basics), AWS (Basics), DBMS, Computer Networks, Data Structures & Algorithms

Internships

ML Intern — Infosys Springboard, India Oct' 2025 – Dec' 2025

- **Led the development of an end-to-end ML system** for loan default prediction using **24+ borrower and loan features**, improving **risk detection accuracy by ~30%**.
- Trained and compared 7 ML models (Logistic Regression, Random Forest, XGBoost, etc.), with XGBoost achieving ~15–20% higher F1-score than baseline models.
- Performed EDA on Kaggle and Microsoft credit datasets (100K+ records) to identify key default drivers such as credit score, DTI, and loan amount.
- Developed a real-time Streamlit web app for instant credit risk prediction, reducing manual underwriting effort by ~40%.
- Improved model generalization by ~18% through cross-dataset validation and production-ready preprocessing pipelines.

Projects

Hospital Load Forecasting Project | ML,Streamlit,Time-Series Forecasting Dec' 2025 – Jan' 2025

- Developed an end-to-end forecasting system that predicts hospital departmental occupancy by modeling anisotropic temporal patterns and latent congestion using unsupervised clustering (DTW + K-Shape) and supervised learning (Random Forest, ARIMA).
- Achieved 23.57% sMAPE (50% better than baselines) and 0.988 F1-score for critical alert detection on synthetic data generated from MIMIC-IV clinical patterns.
- Designed a predictive resource modulation engine that translates forecasts into actionable staffing recommendations
- Built an interactive Streamlit dashboard supporting real-time data upload, manual daily entry, and operational alerts demonstrating real-world deployability.

Smart Waste Management System | ML, Sensors, GPS, Optimization,IOT Integration Jan' 2025 – Feb' 2025

- Built an IoT-enabled smart waste system that auto-segregates dry/wet waste and monitors fill level, methane, GPS, and temperature in real time.
- Triggered cloud alerts via API on bin fullness and optimized collection routes using a shortest-path algorithm (Dijkstra), cutting unnecessary trips.
- Predicted future overflow events with LSTM/Prophet models, enabling proactive scheduling and **~30% reduction in collection trips** for scalable smart city deployment.

Achievements

- Two-seasons AU DevDayT20DSA Challenge Winner conducted by AlgoUniversity. Jan' 2025 – Feb' 2025
- Runner up -Vibecode India Hackathon 2025: Built an Agentic AI application using langchain models. Jun' 2025 – Jun' 2025
- Winner- Anomaly hunter league Hackathon by school of cse,ML,LPU. Jan' 2026 – Feb' 2026

Certifications

- Cloud Computing-NPTEL Jan' 2025 – Apr' 2025
- BuildingAgentic-Ai Applications-programming pathshala Jun' 2025 – Jul' 2025

Education

Lovely Professional University Punjab **2023 – 2027**

Computer Science and Engineering — CGPA: 8.49 *Phagwara, Punjab*

Jawahar Navodaya Vidyalaya,Yanam **2022 – 2023**

12th with Science — Percentage: 80.8% *Yanam, Puducherry*

Jawahar Navodaya Vidyalaya,Yanam **2020 – 2021**

10th with Science — Percentage: 90.2% *Yanam, Puducherry*